

IoT Wireless Weather Monitoring System

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INTRODUCTION

- Automated monitoring of temperature, humidity, and light
- It uses an **ESP32 microcontroller**, sensors (DHT11, LDR), relays, and actuators (fan, bulb).
- The system supports both Manual and Automatic Modes

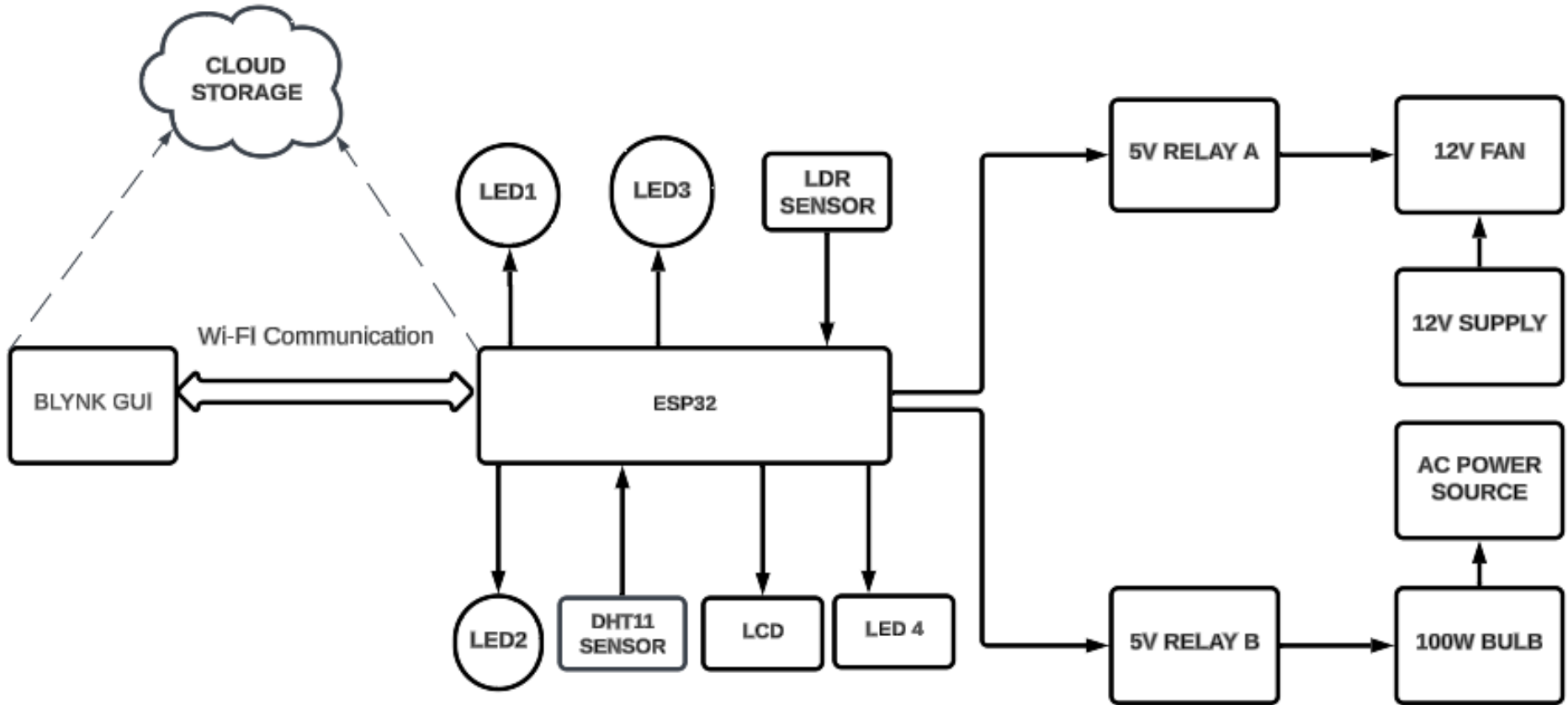
OBJECTIVES

- Maintained automatic temperature control between 23 °C to 28 °C.
- Discovered bidirectional communication between control centre and remote unit.
- Achieved real-time monitoring of the system.

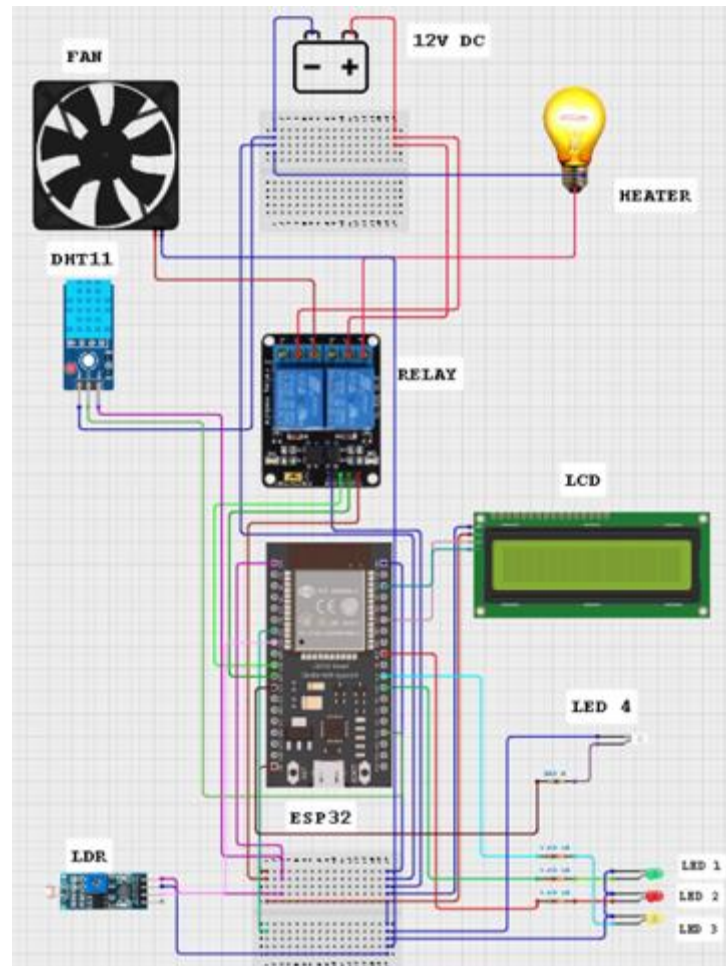
ESPECIFICATINS

- 5 V 2-Channel relay module was controlling fan and heating bulb.
- 3.3 V was powering ESP32 microcontroller, and LDR.
- 12 V power supply was powering 12VDC fan and heating lamp, and AC power supply to power 100W light bulb

BLOCK DIAGRAM

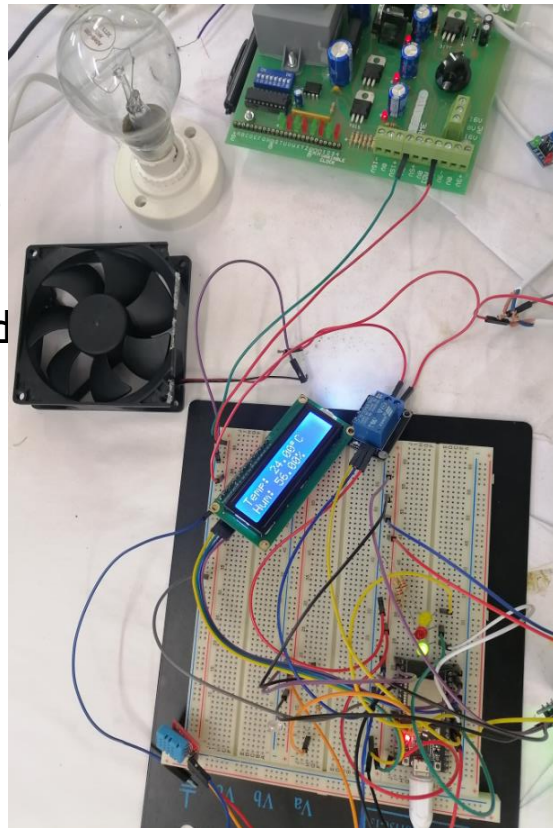


CIRCUIT DIAGRAM

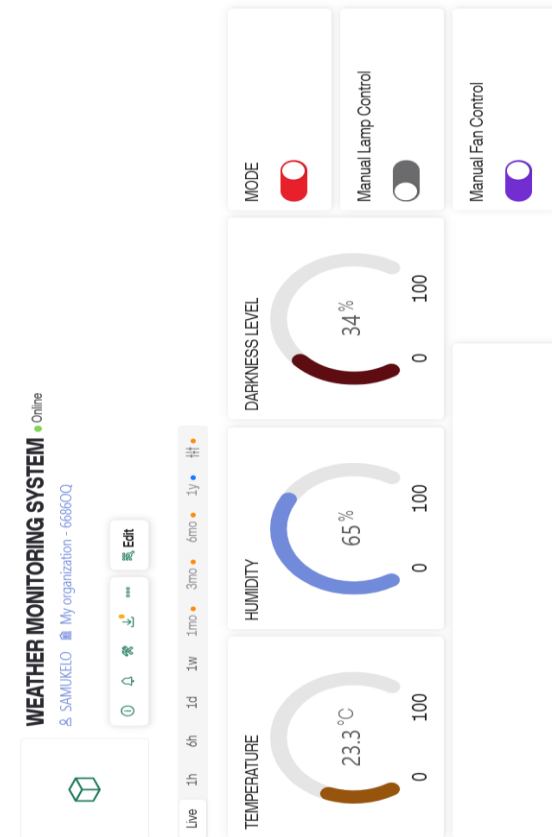


CONSTRUCTION

- Breadboard and component wiring
- LEDs used to indicate temperature and light levels
- Blynk dashboard with switches and gauges
- Manual/Auto control setup using Blynk virtual pins
- Closed housing for safety and neatness



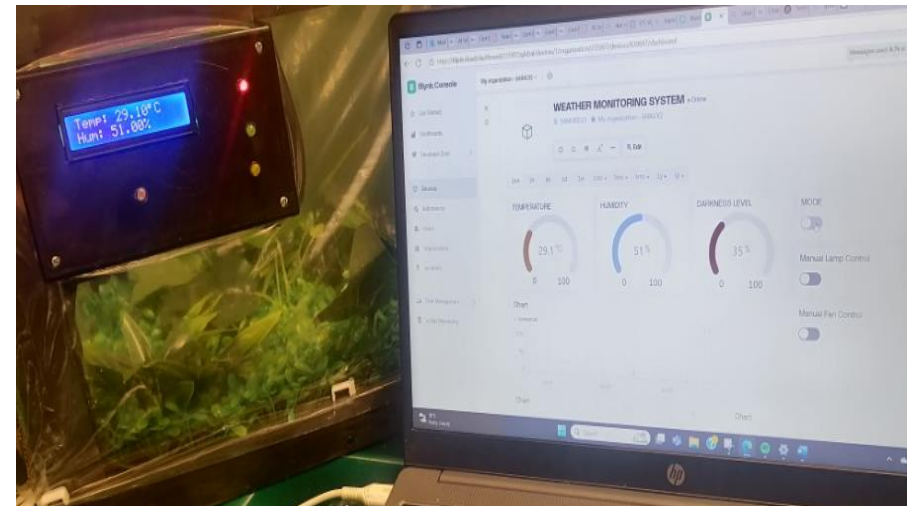
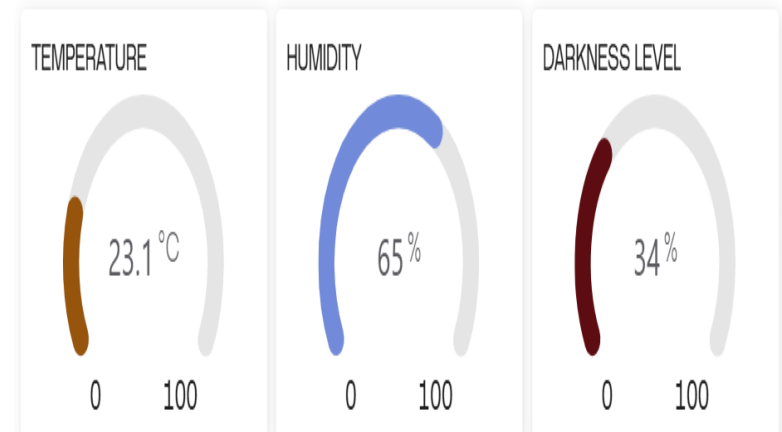
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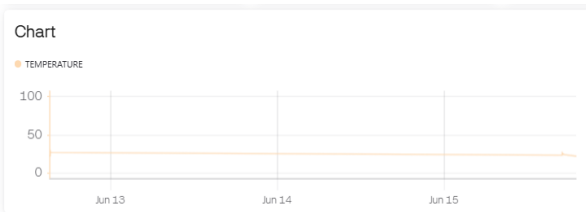
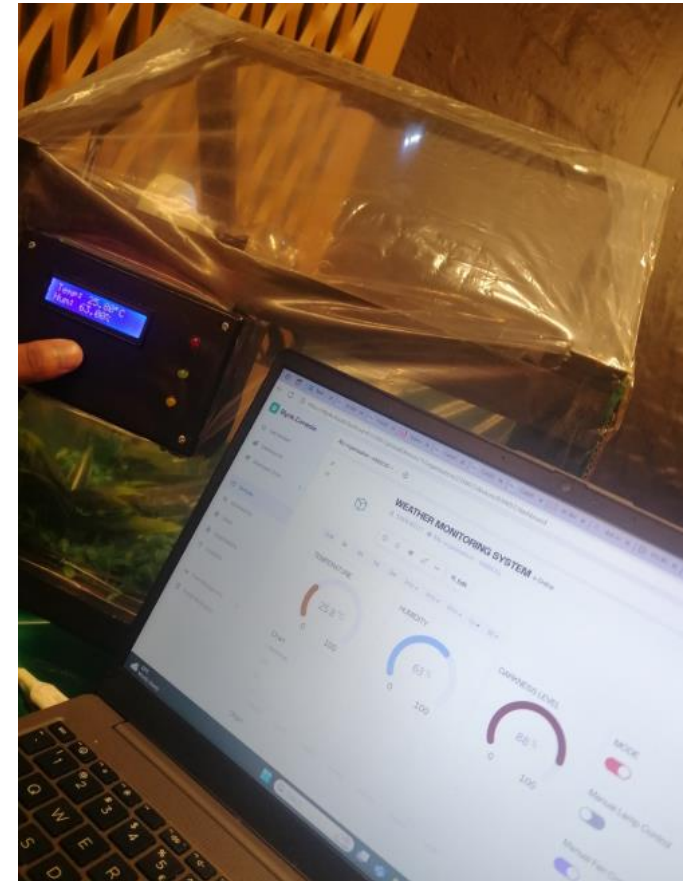
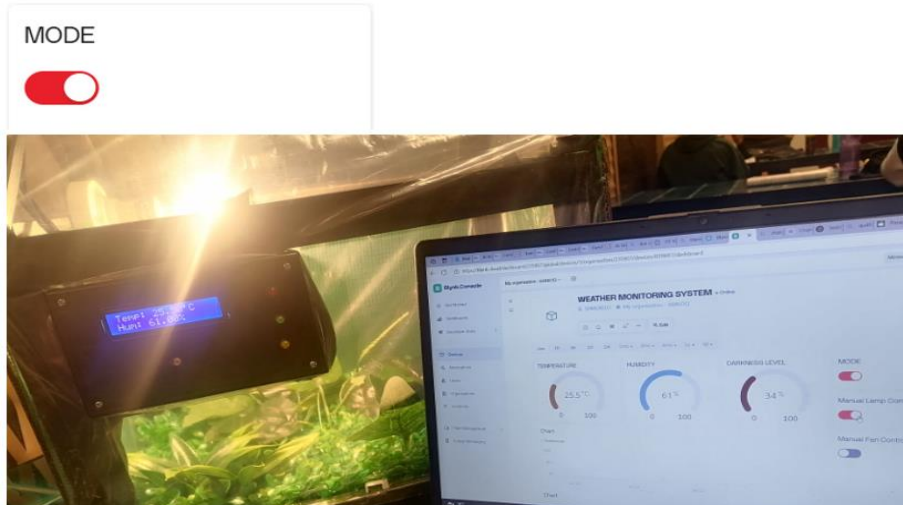
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TESTING

- Sensor readings tested on Blynk gauges (temperature, humidity, light)
- LED indicators tested for below/within/above temperature range
- Fan and bulb triggered based on temperature conditions
- Auto Mode: System responded automatically to sensor inputs
- Wi-Fi connection and Blynk communication tested for stability

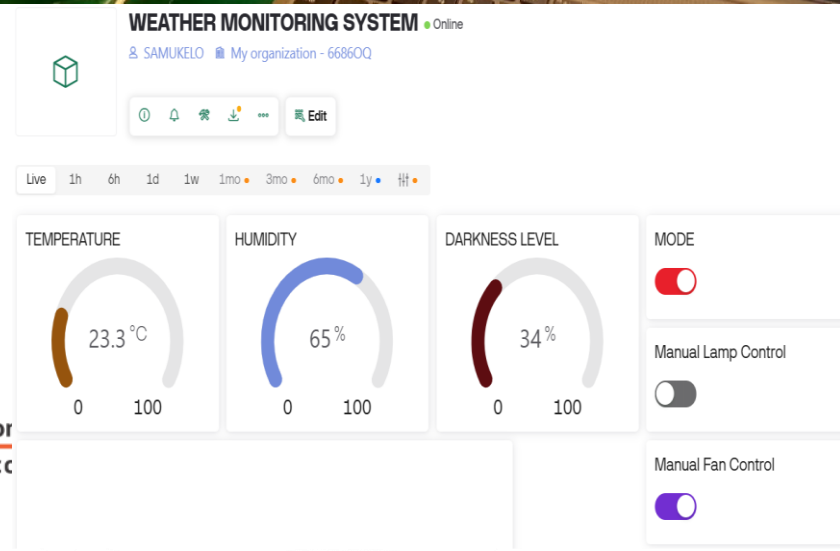


- Darkness detection tested with LDR and LED4
- Manual Mode: Fan and bulb controlled via Blynk switches (V4 & V5)
- History Data checked for proper data logging

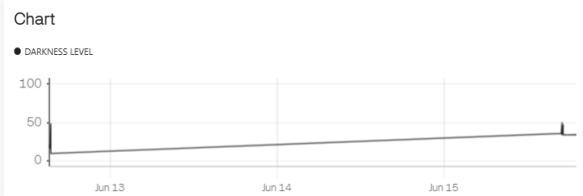
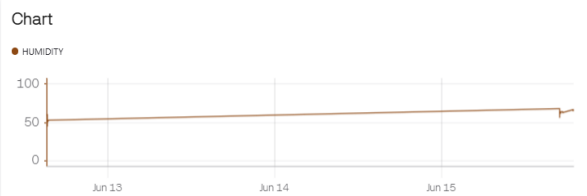
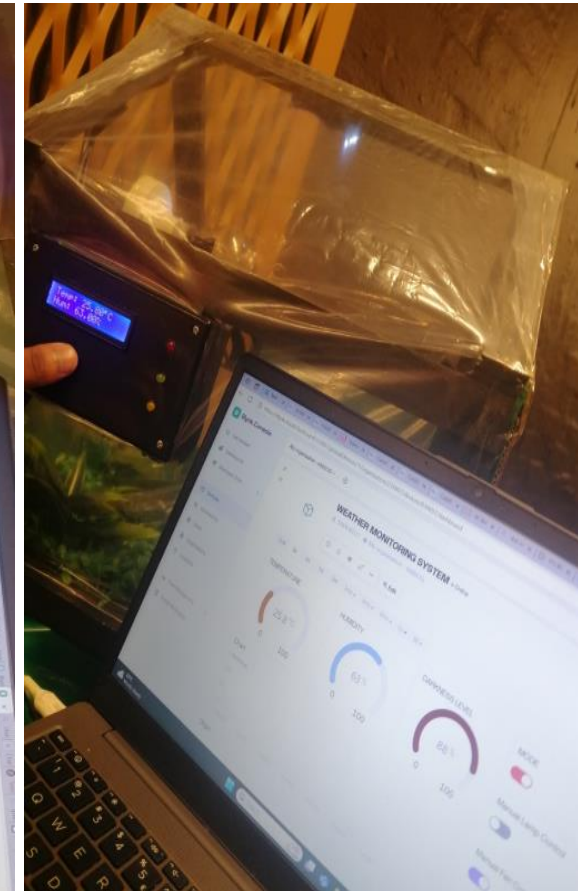
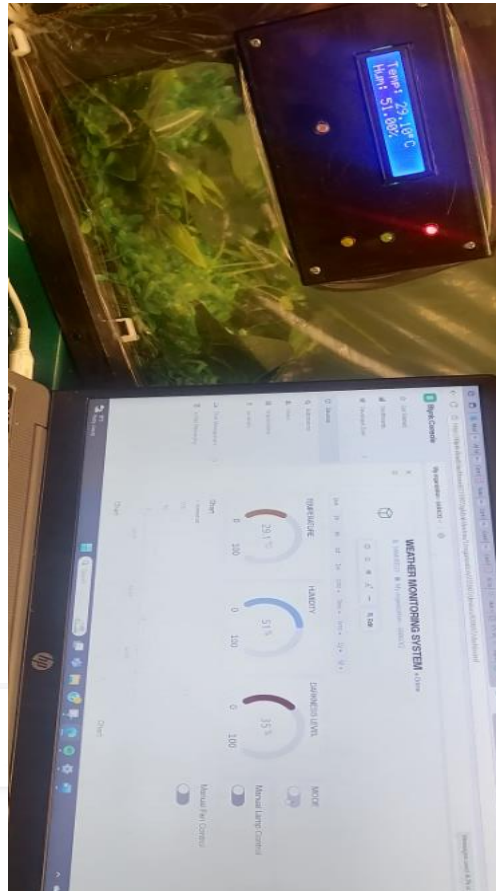


RESULTS

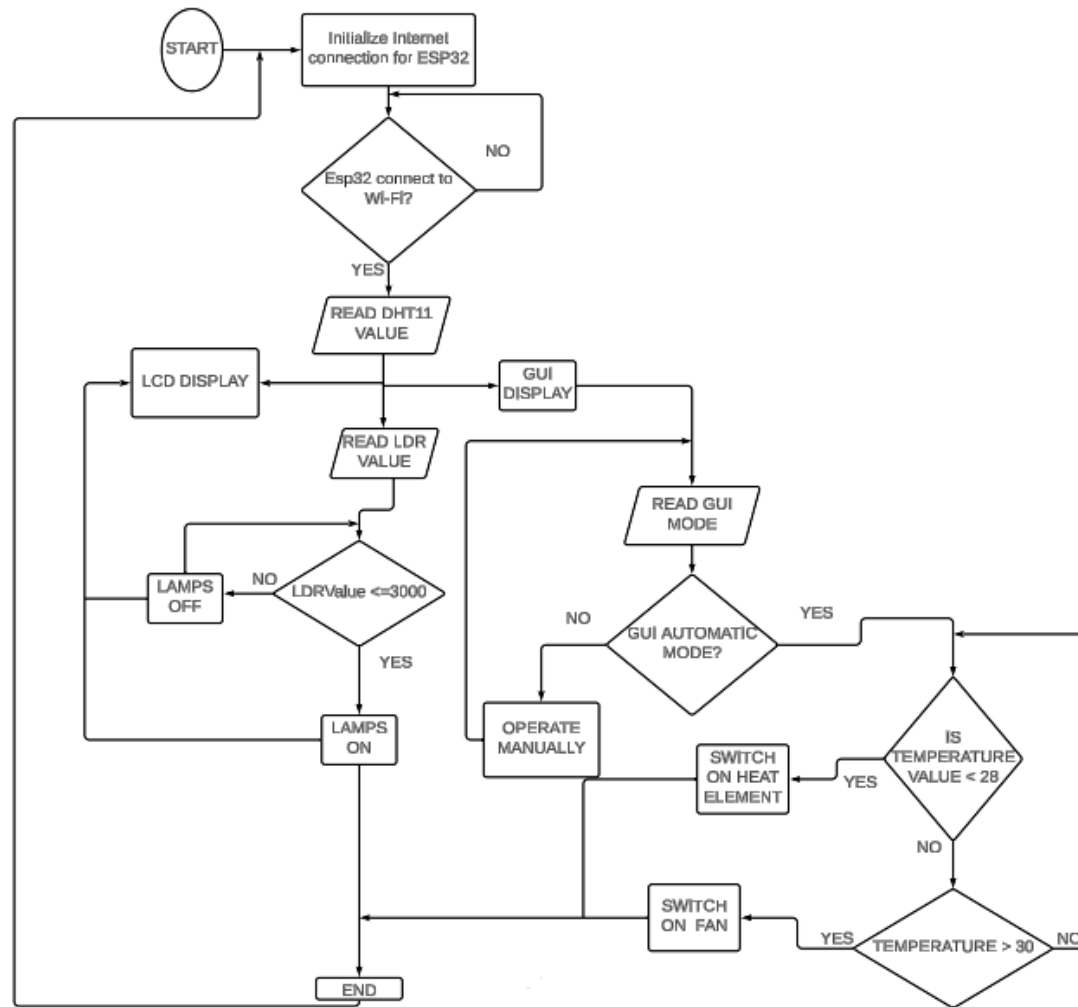
- Temperature, humidity, and light data successfully displayed on Blynk gauges.
- LED1, LED2, LED3 indicated temperature states: below, within, and above range
- Mode switch (V3) toggled between Auto and Manual successfully
- Manual Mode: Fan and bulb controlled successfully via Blynk switches (V4 & V5)



- History Data stored on Blynk Cloud for trend analysis
- LED4 activated in darkness based on LDR input
- Fan turned ON when temperature > 28°C (automatic mode)
- Bulb turned ON when temperature < 23°C (automatic mode)



FLOW CHART



LESSONS LEANT

- Gained hands-on experience in IoT-based system development.
- Learned to integrate sensors (DHT11, LDR) with ESP32 microcontroller.
- Understood relay control for AC and DC devices (fan, bulb)
- Developed skills in using Blynk for real-time monitoring and control

COMPLIANCES

- Complied with to instruction of the supervisor of not using a 12V lamp for a heating element, instead use a 100W light bulb.

PROJECT MANAGEMENT

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THANK YOU

QUESTIONS ?